ContractKV: Supplemental Experimental Record

Bin Zhou and Xi Cao

This supplement provides input-layout measurements, serving controls and detailed experimental outcomes. Protocol labels distinguish qualification granularity, input construction and timing. Source records and aggregation code accompany the tables; section and table numbering is local to this supplement.

S-I. Input Structure and Displacement

We use the frozen input manifests and their measured tokenizer-position ledger to quantify how prompt compilation changes the input. Table S-I distinguishes prepared dataset fields from the controlled serving sequence, including the metadata observation introduced by the runner. It covers all five workload families and every observed serving-order pattern.

Let a_i and b_i be the zero-based token positions of segment i's rendered header marker in raw and candidate chat prompts. We report $d_i = b_i - a_i$, the fraction with changed ordinal index, the fraction with $d_i \neq 0$, mean $|d_i|$, nearest-rank P95 $|d_i|$, and the signed range. Header markers are included; these are not body-only token offsets. Every archived record reports that direct tokenization and tokenization of the rendered chat text agree. The present CPU aggregation checks its arithmetic and completeness, without rerunning model or tokenizer inference. Table S-II does not identify an attention or RoPE causal mechanism.

S-II. Complementary Serving Controls

S-III. Reuse Mechanisms and Serving Breadth

a) CacheBlend interpretation.: The controlled LMCache integration uses 32-token chunks and a separator-matched reference, while ContractKV uses raw order on its own runtime. The paired 0.03 statistic and these configurations differ from CacheBlend's published aggregate-quality evaluation. The adapter's positive mean-score change coexists with individual decreases; this is a controlled quality comparison, not a reproduction of the authors' performance configuration or a causal cross-stack risk ratio.

S-IV. Extended Evaluation Record

We evaluate whether ContractKV (1) obtains complete-cost gains on task-preserving inputs, (2) maintains observed free-running behavior within admitted scopes, (3) adds publication control beyond a preselected layout, and (4) converts template sharing into measurable physical reuse. The primary comparisons use query-qualified execution. Deployment-qualified and bounded-input studies supply separately identified amortization, scale, and diagnostic evidence.

A. Workloads, Policies, and Statistical Units

The workloads derive from RULER [6], HotpotQA [7], NarrativeQA [8], Qasper [9], and RepoBench [10]. The reading-comprehension inputs use the LongBench task collections [11]. The main task-preserving cohort contains three new HotpotQA sources and three new repository sources, selected by a frozen hash order after excluding 567 previously used sources. Each source defines a template with ten tenants and ten rounds. Three query wrappers recur across rounds, with private metadata changing under the declared observation schema. This trace prescribes an authorized query directory whose reuse can be measured explicitly.

The models are Qwen3-8B [12], Mistral-7B-Instruct-v0.3 [13], and Llama-3.1-8B-Instruct [14]. Qwen3-14B is additionally evaluated in the deployment cost study. Temperature is zero, the seed is fixed, and the free-generation budgets are 128 tokens for RULER/HotpotQA, 256 for NarrativeQA/Qasper, and 384 for RepoBench. Longer-output experiments have separate declared budgets and schemas. The primary task-preserving KV pools contain 31,040, 56,800, and 38,704 tokens for Qwen, Mistral, and Llama, respectively.

Raw uses the original prompt order and native APC. *ContractKV* uses the same task content, frozen candidate policy, query-scope eligibility, token/EOS floors, output contract, and service clock. Mistral and Llama retain the previously frozen RepoBench L0 route. A separate cost-adapted variant changes only the cost profile and waiting-budget gate. In the deployment study, D1 uses an already qualified frozen layout without online registration, while D2 adds in-engine verification and transactional registration to that deployment route.

For every request pair, we retain actual prompt/output IDs, task scores, route, qualification identity, timing, and cache counters. LongBench-derived QA uses the frozen token-F1 scorer. RepoBench uses the pre-existing SequenceMatcher-based edit-similarity adapter, while RULER uses its frozen substring-match criterion. We report upward and downward task-score changes using the preregistered 0.03 material-change threshold, and evaluate exact output-ID agreement separately. RepoBench results therefore describe this declared next-line task adaptation, rather than a reproduction of every official evaluation component.

Each primary model has 1,620 future-request pairs across three repetitions, giving 4,860 future pairs and 10,800 total business generations. The six source templates are the independent content units; tenants and rounds are nested repetitions. Reported min/max throughput ranges summarize process repetitions, while confidence intervals resample source-template clusters and repetitions. Separate cohorts and comparisons sharing a raw control are not pooled as independent trials.

TABLE S-I
FROZEN FIVE-FAMILY INPUT ORDER BEFORE CALC. PREPARED FIELDS COME FROM THE ARCHIVED SESSION INPUTS; SERVING ORDER IS RECONSTRUCTED FROM EVERY RECORDED RAW-ORDER INDEX. REPEATED SUPERSCRIPTS DENOTE CONTIGUOUS COMPILER UNITS, NOT EXTRA DOCUMENTS.

Family	Prepared task fields	Rendered serving segments
RULER	SYS→SYN→QUERY	SYS→OBS→SYN^{16}→QUERY
HotpotQA	SYS→RAG→QUERY	SYS→OBS→RAG^{16}→QUERY
NarrativeQA	SYS→RAG→QUERY	SYS→OBS→RAG^{16}→QUERY
Qasper	SYS→RAG→QUERY	SYS→OBS→RAG^{16}→QUERY
RepoBench	REPO→CODE→QUERY	SYS→OBS→REPO^{16}→CODE^{k} →QUERY

SYS, OBS and QUERY denote SYS_PROMPT, TOOL_OBS and USER_QUERY. RAG, SYN, REPO and CODE denote RAG_CHUNK, SYNTHETIC_CONTEXT, REPO_CONTEXT and CODE_PREFIX. For RepoBench, $k \in \{1, 6, 16\}$; the runner adds a SYS segment even when the prepared task has no system body. Outer chat-control tokens are outside this segment-order view. The metadata-only OBS is a controlled runner field, not a claim about production prevalence.

TABLE S-II
INPUT DISPLACEMENT FROM 4,695 ARCHIVED SEGMENT OBSERVATIONS IN 240 MODEL/SESSION INPUTS (THREE TOKENIZERS OVER 80 SESSION INPUTS). CANDIDATE LEVELS FOLLOW THE FROZEN PER-MODEL POLICY, INCLUDING L0, AND ARE NOT CONDITIONED ON VERIFIER SUCCESS. THESE NESTED OBSERVATIONS ARE NOT 4,695 INDEPENDENT TASK SOURCES. PERCENTAGES AND MEANS ARE SEGMENT-WEIGHTED.

Family	Segments	Ordinal moved (%)	Token moved (%)	Mean $\lvert d_i \rvert$	P95 $\lvert d_i \rvert$	Signed range
RULER	570	89.5	89.5	114.31	1243	[-65, 1471]
HotpotQA	2,280	73.6	73.6	141.45	1593	[-66, 3118]
NarrativeQA	570	82.6	82.6	153.56	1479	[-64, 3030]
Qasper	570	29.8	29.8	52.21	49	[-49, 2377]
RepoBench	705	30.5	30.5	70.27	49	[-49, 4449]
ALL	4,695	64.9	64.9	118.10	66	[-66, 4449]

TABLE S-III
DEPLOYMENT-TEMPLATE DENSITY REPLAY FROM 10 TO 80 REGISTERED IDENTITIES, THREE FRESH PROCESSES PER POLICY/POINT. THE CLASS-SPECIFIC SLOS ARE 750/1,500/3,000 MS; THE POOL HAS 2,386 BLOCKS. BLOCK COLUMNS AVERAGE PROCESS PEAKS. CLIENT CONCURRENCY IS FOUR AND SERVER MAX-NUM-SEQS IS EIGHT. IDENTITIES ARE NOT SIMULTANEOUS GPU REQUESTS.

IDs	Raw req/s	CKV req/s	Raw blocks	CKV blocks
10	0.900	9.408	525.67	146.00
20	1.032	10.781	530.00	149.00
30	1.059	10.532	530.00	149.00
40	1.073	11.057	530.00	149.00
60	1.112	11.072	530.00	149.00
80	1.152	11.033	530.00	149.00

TABLE S-IV
SEPARATE BOUNDED-INPUT RULER AUDIT. COLD AND FAN-OUT TRACES ARE POOLED ONLY WITHIN THIS DECLARED PROTOCOL, NOT WITH THE FAITHFUL 16K HOTPOTQA/REPOBENCH COHORT.

Model	Cold C/N	Fan-out C/N	Total C/N
Qwen	210/270	516/540	726/810
Mistral	210/270	516/540	726/810
Llama	210/270	516/540	726/810

TABLE S-V
COMPLEMENTARY EXECUTION-CLOCK SLO STATISTICS FOR THE FOUR-CONFIGURATION DEPLOYMENT COHORT. THESE RATIOS USE ARCHIVED BUSINESS-BATCH EXECUTION WALL, NOT COMPLETE QUALIFICATION-CHARGED SERVICE WALL. DEADLINES ARE CONFIGURATION-SPECIFIC AND MATCHED WITHIN EACH RAW/CONTRACTKV PAIR. NET TTFT IN THE MAIN TABLE SEPARATELY ADDS RESIDUAL VERIFICATION EXPOSURE.

Configuration	SLO multiplier	Slack (s)
Qwen3-8B	2.832×	2
Qwen3-14B TP=2	2.916×	2
Mistral-7B	1.261×	2
Llama-3.1-8B	1.189×	6

B. Complete-Cost Performance on Preserved Tasks

Table S-IX shows complete-cost throughput gains on all three models. Qwen achieves 1.619× throughput and a 43.87% reduction in mean readiness TTFT. The 1.615–1.623× range across fresh processes indicates a repeatable execution advantage on the selected cohort. Mistral and Llama obtain smaller gains of 1.109× and 1.070×, with lower admitted coverage. All three obtain compiled HotpotQA responses: 516, 342, and 270 future pairs, respectively. Qwen additionally admits 168 RepoBench future pairs.

The physical workload changes explain how the improvement is obtained. Table S-X separately reports business and qualification prefill. Qwen reduces business new-prefill work by 46.67%; including qualification still leaves approximately 45.44% less measured prefill than raw. The corresponding complete-cost throughput is lower than a prefill-only ratio because decoding, qualification, and control work remain on the service path.

A second independent cohort, used for the three-way cost-policy study, contains three different HotpotQA sources and three different repository sources. The unchanged query-qualified ContractKV path achieves 1.803× complete-cost throughput on Qwen, with a three-run range of 1.798–1.808× and 774/1,620 compiled future pairs. This supplies a second source-disjoint observation of substantial full-cost benefit; it is reported separately from the three-model cohort.

TABLE S-VI
COMPARISON OF PUBLISHED MECHANISMS AND THEIR ROLE IN THIS STUDY. THE TABLE COMPARES RESPONSIBILITIES AND EVALUATED EVIDENCE, NOT UNREPORTED SECURITY CAPABILITIES OF THE OTHER SYSTEMS.

System	Published mechanism / reuse unit	Relationship to the present evaluation
SGLang [1]	RadixAttention organizes exact-prefix KV reuse for structured LLM programs.	A native reuse substrate; the separate frozen-layout transfer test does not implement ContractKV certification or PSTP.
Prompt Cache [2]	A schema defines reusable prompt modules and their positional treatment.	Modular reuse design; discussed as a distinct way to expose reusable attention state.
CacheBlend [3]	Cached passages are fused with selective recomputation of token KV.	Matched-native paired task outcomes are measured; cross-stack speeds are not ranked.
EPIC [4]	Position-independent caching with LegoLink to address document-boundary attention effects.	A different execution mechanism; no direct EPIC performance reproduction is claimed.
Parrot [5]	Semantic Variables expose application input/output dataflow to the serving system.	Application dependency information motivates service-aware control; its optimization target is application execution.
ContractKV	A dependency-valid stable-prefix layout is bound to behavioral qualification, authorization/lifecycle identity and a transactional publication boundary.	Same-engine admission/publication controls, template amortization, free-running task pairs and physical reuse are measured separately.

TABLE S-VII
FOUR SEALED OPERATING POINTS UNDER THE DEVELOPMENT-FROZEN BENEFIT GATE. NET BENEFIT IS PER CERTIFICATE; REJECTED POINTS SERVE RAW. CANDIDATE DIAGNOSTICS WERE ALSO RUN FOR REJECTED POINTS, SO THIS TABLE DOES NOT MEASURE AVOIDED ONLINE VERIFIER CALLS.

Prefix	Uses	Slack (s)	Route	Net benefit (s)
1500	8	3.5	Raw	0.00
3000	12	0.5	CKV	7.04
5000	10	1.5	CKV	10.84
3500	6	6	Raw	0.00

Mean improvements coexist with higher p99 values in Table S-IX. The source-cluster/repetition 95% intervals for percentage p99 reduction are [-56.247, 26.802], [-60.576, -0.137], and [-16.784, 0.151] for Qwen, Mistral, and Llama. The Mistral interval is entirely on the side of increased latency. For mean readiness reduction, the corresponding intervals are [5.640, 79.479], [-0.350, 34.781], and [1.550, 5.724]. The corresponding tradeoff is higher aggregate throughput with nonuniform tail-latency effects.

C. Input Fidelity as a Controlled Factor

We isolate task-input construction with a four-cell Qwen experiment on two observed HotpotQA and two observed RepoBench sources: fixed-cap/raw, preserved-input/ContractKV, preserved-input/raw, and fixed-cap/ContractKV. All four are new processes on the same GPU, with the same 16K window, two-request concurrency, policy, scope, margin, scoring, and clock. Two sources retain their full input and two crop optional context by the frozen rule.

Restoring task structure increases observed coverage from 8.33% to 31.11% without changing the acceptance rules (Table S-XI). In the preserved-input condition, all 360 future raw/ContractKV outputs have identical IDs and scores. The rise in absolute task scores is an input effect: both routes receive the same retained content. The ablation jointly changes packing and contiguous segment rendering; the later independent cohort tests the resulting pipeline as a whole.

The bounded-input audit explains why this factor matters. Under the older word-limited adapter, eight of ten audited HotpotQA sources lose a reference-answer literal present in the stored context. Five of ten RepoBench code prefixes are empty after budgeting, and whitespace is flattened in all ten. Literal availability alone is not semantic answerability, but the code-prefix and formatting losses directly change what the model is asked to complete. The retained historical controls and the new input protocol are therefore kept as distinct experiments.

D. Free-Running Behavior and Qualification Coverage

All 1,296 admitted compiled future pairs in the primary cohort have identical output IDs and unchanged task scores. Including initial business rounds gives 1,440 such compiled pairs. Table S-XII reports exact agreement together with absolute reference utility and full-route score changes.

All material full-route changes occur on L0. The three Llama decreases occur on future requests; the Mistral decrease occurs on an initial business request. The corresponding output variants also occur in the scheduled raw-only repeats. These controls demonstrate that raw execution itself varies across the recorded runs, but do not establish that every L0 difference is independent of policy-associated execution conditions.

Absolute quality remains visible. Llama has zero-F1 raw references for two of the three HotpotQA sources and preserves them in 180 compiled future pairs; Mistral has 84 compiled future pairs from a zero-F1 reference source. A positive F1 or edit score is also not equivalent to complete semantic correctness. The cache-layer result is the observed preservation of the selected reference behavior, with absolute task scores reported alongside it.

The broader, five-family query-qualified cohort supplies complementary coverage evidence. Across 36 cells and 12,150 future raw/strategy pairs, its 2,442 admitted compiled pairs have no ID or score changes. Full-route material decreases total 33 and occur on L0. That cohort uses fixed-cap packing and a different source set, so it remains separate from Table S-XII.

TABLE S-VIII
MATCHED-NATIVE PAIRED TASK-SCORE DISTRIBUTIONS (2,250 PAIRS/METHOD). INCREASES AND DECREASES USE THE DECLARED 0.03 THRESHOLD. EACH ROW RETAINS ITS OWN BASELINE AND PROMPT/RUNTIME CONFIGURATION; THESE MEANS ARE NOT A COMMON CROSS-STACK ACCURACY OR SPEED COMPARISON.

Method	Matched reference	Reference mean	Method mean	Increases	Decreases
CacheBlend adaptation	cacheblend_raw	0.333210	0.339817	331	224
ContractKV	vllm_apc	0.353780	0.353744	0	2
Manual layout control	vllm_apc	0.353780	0.356323	60	39
SGLang frozen layout	sglang_raw	0.353739	0.353874	2	2

TABLE S-IX
PRIMARY TASK-PRESERVING, QUERY-QUALIFIED EVALUATION: SIX SOURCE-DISJOINT TEMPLATES, THREE FRESH-PROCESS REPETITIONS, ZERO SLACK. THROUGHPUT USES ALL BUSINESS REQUESTS AND COMPLETE SERVICE WALL; COVERAGE AND READINESS STATISTICS USE FUTURE REQUESTS. THE THREE-RUN RANGE IS MIN/MAX, NOT A CONFIDENCE INTERVAL.

Model	Compiled / future pairs	Full-wall throughput	Three-run range	Readiness TTFT, raw → ContractKV (ms) Mean	p99	Mean reduction (%)
Qwen3-8B	684 / 1,620	1.619×	1.615–1.623	4,443.3 → 2,494.2	8,561.8 → 9,146.6	43.87
Mistral-7B	342 / 1,620	1.109×	1.107–1.110	5,158.2 → 4,534.7	9,418.2 → 9,511.3	12.09
Llama-3.1-8B	270 / 1,620	1.070×	1.068–1.072	4,194.7 → 4,027.3	7,851.3 → 7,905.7	3.99

TABLE S-X
ACTUAL NEW-PREFILL TOKENS PER REPETITION IN THE PRIMARY COHORT. QUALIFICATION IS A SEPARATE WORKLOAD, NOT SUBTRACTED FROM MEASURED SERVICE TIME.

Model	Raw business	CKV business	Qualification
Qwen	6,008,336	3,204,304	73,580
Mistral	7,074,492	5,986,864	51,760
Llama	5,822,228	5,149,118	38,807

TABLE S-XI
SAME-SOURCE INPUT-CONSTRUCTION ABLATION. EACH INPUT CONDITION HAS 360 FUTURE RAW/CONTRACTKV PAIRS; FULL WALL INCLUDES 400 BUSINESS RESPONSES PER PATH. ONE PROCESS PER CONDITION.

Metric	Fixed cap	Preserved
HotpotQA raw mean F1	0	0.22222
RepoBench raw mean task score	0.12067	0.36667
Compiled future pairs	30 / 360	112 / 360
Compiled ID differences	0 / 30	0 / 112
Full-wall throughput/raw	1.039×	1.117×
GPU qualification calls	3	9

1) Why evidence must be scoped: An earlier first-round qualification protocol admitted later changing wrappers and produced 108 material decreases among 3,672 compiled pairs. Query binding reduced the exposure but did not by itself settle metadata sensitivity: a later Qwen NarrativeQA source changed its raw output when only a permitted turn field changed, while its compiled output retained the witness. The candidate minimum margin was 0.40625, above the 0.125 threshold. Cold/warm and tested execution-shape controls did not eliminate that behavior. This was a counterexample within the earlier declared empirical scope, and motivated the final eligibility rule in the main paper's certification section. The artifact retains every intermediate outcome; the cohorts are not treated as a matched risk-reduction curve.

TABLE S-XII
BEHAVIORAL OUTCOMES IN THE PRIMARY COHORT. POSITIVE-SCORE COUNTS CONCERN THE COMPILED FUTURE SUBSET. FULL-ROUTE CHANGES INCLUDE INITIAL AND FUTURE BUSINESS ROUNDS AND USE $|\Delta s| > 0.03$.

Metric	Qwen	Mistral	Llama
Compiled future pairs	684	342	270
Compiled ID differences	0	0	0
Compiled any score decrease	0	0	0
Raw score > 0 in compiled subset	684	258	90
Full-route score increases	0	2	5
Full-route score decreases	0	1	3

2) Certified output horizon: Separate long-output conditions verify complete natural-stop witnesses and compare subsequent free-running responses. Qwen produces 642 tokens for a 128-item schema, Mistral 769 tokens for a 128-item schema, and Llama 578 tokens for a separately declared 192-item schema. Each has six correct future raw/compiled pairs across three fresh processes. The Llama 224-item condition omits the final 100 items and is retained as a task-completion failure. The horizon evidence is thus based on actual generated and verified sequences, not an output-budget setting alone.

E. What Certification Adds Beyond Layout Reordering

D1 isolates the performance of a prequalified frozen layout; D2 adds registration, verification, and publication governance on that deployment path. In the same-engine comparison, absolute 500-ms SLO goodput is 1.472 requests/s for D1 and 1.428 requests/s for D2, with nearly equal mean TTFT (194.95 and 194.72 ms). D2 preserves most of the available layout benefit while executing the governance path.

The complementary test is a candidate that should not be published. In two separate 20-case controlled sets, all ten unsafe layouts in each set are rejected and all rejected transactions publish zero candidate tokens. A 14B control additionally checks real post-decision responses against raw task scores.

TABLE S-XIII
INDEPENDENT ZERO-SLACK REGISTRATION-COST SUPPLEMENT, THREE PROCESSES/CONFIGURATION. NET MEAN TTFT CHARGES MEASURED DIAGNOSTIC REGISTRATION AMORTIZED OVER FUTURE REQUESTS. POSITIVE VALUES INDICATE REDUCTION VERSUS THIS COHORT'S RAW REFERENCE. THESE ROWS ARE NOT PAIRED SLACK SETTINGS OF THE MAIN DEPLOYMENT TABLE.

Configuration	Net mean TTFT reduction
Qwen3-8B	17.50%
Mistral-7B	−2.84%
Llama-3.1-8B	7.69%
Qwen3-14B, TP=2	10.59%

These cases demonstrate the value missing from a layout-only performance number: behaviorally unsupported candidates do not acquire shared-cache visibility through the registration path.

The lifecycle suite records live source-hook, concurrency, cancellation, duplicate and timeout controls. Binding perturbation tests separately check request-side client functions on CPU. The 325,000-operation stress study exercises the in-memory transactional state machine; it is not counted as GPU service requests. The empirical behavioral test and the transactional publication property remain separately attributable components of the system.

F. Template Amortization and Deployment Economics

The deployment-qualified study isolates the granularity of registration while holding the layout and locality fixed. At 25-tenant locality and its frozen 80% load point, session registration issues 125 verifier calls and template registration five. Fan-out rises from 9 to 225 requests per registered object, and unhidden verification exposure falls from 25.05 to 3.24 ms/request. Prefix-hit ratios remain close (97.82% and 98.19%), with mean TTFT 64.02 and 62.91 ms. The dominant change is where the registration cost is paid and shared, rather than a different physical layout.

These measurements use deployment-qualified registration. In the query-qualified studies, the analogous accounting unit is a selected query scope shared by authorized tenants. For example, the second Qwen cohort obtains 774 compiled future pairs from 27 issued scope-repeat decisions and 42 total GPU qualification calls, including failed attempts. The two protocols support the same service-level principle—pay model-backed qualification at a reusable object boundary—with different measured decision counts.

The four-configuration deployment evaluation reports 29.41–44.69% lower net mean TTFT under residual-verification accounting with 2s/6s slack. Table S-XIII is a separate diagnostic-registration cohort. Its net mean is the observed business TTFT plus measured total registration-controller cost divided by future requests, with no slack subtraction. The 14B condition uses a 96-token budget; the other rows use their frozen family budgets. This is amortized mean accounting, not a live per-request p95/p99 overlap measurement or a matched zero-slack recomputation of the deployment table.

A separate common-scope HotpotQA study uses the same 30 sessions across three models and three repetitions, producing 2,430 pairs. Business prefill reductions are 48.05% for Qwen, 40.69% for Mistral, and 38.38% for Llama. Its zero-slack macro mean TTFT improvement is 5.242%, with a source-cluster 95% interval of [-2.162, 12.564]%. The primary preserved-input experiment adds independent evidence under the structure-preserving input protocol.

TABLE S-XIV
INDEPENDENT SECOND-COHORT QWEN COMPARISON. BOTH STRATEGIES SHARE THE SAME RAW CONTROLS. THREE FRESH-PROCESS REPETITIONS; FULL WALL INCLUDES ALL BUSINESS RESPONSES.

Metric	Original gate	Cost-adapted
Compiled future pairs / 1,620	774	258
GPU qualification calls	42	12
Issued scope-repeat decisions	27	9
Full-wall throughput/raw	1.803×	1.070×
Readiness p99 (ms)	13,386.9	8,801.9
Full-route material decreases	8	6
Compiled ID differences	0	0

The deployment service study also measures fixed-arrival SLO behavior and allocator occupancy. At its 25-tenant, 80%-load point, 500-ms goodput increases from 0.173 to 0.831 requests/s. At 80 registered tenants, peak used KV blocks fall from 530 to 149 in a 2,386-block pool. These are deployment-qualified, marker-bearing service measurements, paired with a separate task-quality audit. They describe a specific load and cache configuration; the primary query-qualified throughput results in Table S-IX use a different, task-bearing closed-loop protocol.

G. The Cost of Controlling Qualification Waiting

Long inputs expose a specific cost-model failure. In a development RepoBench source, the predicted conditional GPU verification cost is 2.027 s, while the measured GPU RPC takes 4.034 s and the full controller takes 4.142 s. The often lower pre-reference expected cost also includes eligibility weighting and is not the same quantity. Synchronous unsuccessful registration enters the next business wave's readiness delay.

We evaluate one frozen cost-adapted variant: a length-bucketed cold-cost envelope and an incremental fan-out waiting budget, with the existing benefit cap, scope, margins, and publication mechanism retained. On the four-source development control it preserves 112/360 compiled future pairs, reduces GPU calls from nine to six, and lowers RepoBench p99 from 10.135 to 6.171 s. Its full wall falls from 995.669 to 983.412 s.

Independent results show a different tradeoff (Table S-XIV). The cost profile's HotpotQA support ends at an 8,192-token bucket, with actual observations only up to approximately 6,880 tokens; the new 13.9–14.6K-token candidates fall outside that support. Its frozen rule consequently rejects 27 HotpotQA scope-repeat decisions, and the waiting budget rejects another nine RepoBench decisions. Eighteen rejected decisions correspond to qualifications the original gate actually issues. This is a loss of beneficial coverage, not a behaviorally necessary rejection.

The variant's global p99 point falls by 34.25%, but the direct source-cluster/repetition reduction interval [-12.145, 37.569]% crosses zero; throughput falls to 59.35% of the original gate.

TABLE S-XV
KNOWN-SOURCE RESIDENCY PROBES. MISSING HITS HAVE A RECORDED DELETION OF THE LAST MATCHING NATIVE BLOCK MAPPING BEFORE LOOKUP. PROBE TIMING IS EXCLUDED FROM PERFORMANCE COMPARISONS.

Model	Compiled	Full	Zero / partial	Missing with trace
Llama	20	14	6 / 0	6 / 6
Mistral	16	10	6 / 0	6 / 6
Qwen	26	25	0 / 1	1 / 1

We therefore retain the original query-qualified gate as the main measured configuration. All eight/six material decreases remain in the full-route results and occur on L0. Some variants appear in raw repeats; five strategy/position variants are not observed in the three raw controls, so their cause remains unresolved.

H. From Published Prefixes to Actual Residency

The primary cohort contains 189 incomplete-hit compiled business requests: 135 Llama zero hits, 42 Mistral zero hits, and 12 Qwen partial hits. All 1,440 compiled business requests match their certified prefix IDs and have a positive, block-aligned commit length. A complete CPU audit joins 5,400 strategy business requests to their qualification, scheduler, and runtime records; every incomplete-hit request has a later full-hit peer in its two-request wave. This rules out the inference that an entire turn of consecutive tenants necessarily missed.

We then freeze three known-source mechanism probes, one per model. They replay archived prompt IDs, witnesses, candidate choices, and namespaces, with immediate serial/concurrent controls and reuse after intervening raw work. Read-only instrumentation records actual lookup keys, all matching block mappings, publication, release, and native reclamation. The three cells execute 138 business requests and four successful qualifications.

All 13 incomplete hits have matching backend salt and certified hash chains, an absent mapping immediately before lookup, and a preceding native eviction of the hash's last mapping with reference count zero. Immediate post-commit reuse succeeds in all 12 controls. In the Llama example, an intervening raw request reclaims the committed prefix's first block; the following compiled request finds no mapping, recomputes, and enables its peer to obtain a full hit. Qwen's partial hit begins at block 365 rather than at the first block. These events are consistent with the native free-queue reclamation design [15].

The probes establish the mechanism on the three replayed sources; historical logs lack the full eviction trace needed to attribute every earlier event individually. They introduce no production-policy modification and no new performance claim. The throughput in Table S-IX already contains the actual residency misses and recomputation costs.

I. Cross-Position Baselines and Quality Interpretation

Our matched-native quality study retains each system's own raw execution as its reference. The four methods supply 9,000 pairs in total, 2,250 per method. For the two principal comparisons, CacheBlend records 224 material decreases and 331 increases, with mean score changing from 0.333210 to 0.339817. ContractKV records two decreases and no increases, with mean score changing from 0.353780 to 0.353744. The two baseline means, runtimes, prompt construction, and separator configurations differ.

These observations show why mean quality and per-request preservation answer different questions: a method can improve average score while changing some individual answers adversely. We report both directions and threshold sensitivity, and use the same-runtime D1/D2 experiments for performance attribution. The CacheBlend comparison is not used as a cross-stack speed ranking or a multiplicative causal safety advantage. Its published cached-fusion mechanism targets an accuracy/latency tradeoff [3]; the present paired-regression threshold and controlled workloads measure a different aspect of behavior.

S-V. SOURCE-LEVEL OUTCOMES AND REGISTRATION BOUNDARIES

Table S-XVI retains every source from the frozen primary cohort. No source or score was filtered after generation. It separates absolute reference utility, all-route score changes, compiled coverage and source-level tails. The common denominator does not make tenant or process observations independent content sources.

a) Source identities and input status.: H1/H2/H3 are HotpotQA documents 109/19/69, respectively. H1 retains full text; H2/H3 crop only optional context. R1/R2/R3 are the RepoBench Python v1.1 repositories `Gene-Weaver/VoucherVision`, `OSU-NLP-Group/SeeAct`, `sinoyou/nelf-pro`, respectively, all retained in full. Canonical source keys and input hashes accompany the frozen record.

A. Registration Trust Boundary

Table S-XVII lists every frozen helper case. Identity, role, authentication, hash and declared-publication checks cannot infer whether trusted shared-content declarations are semantically correct. A valid hash/tag on a private tool response mislabeled shared remains an operator error. The live endpoint and PSTP controls are separate evidence, not an end-to-end disclosure detector.

B. Open-Arrival Uncertainty

Table S-XVIII gives the absolute goodput increment at the main figure's deployment operating point. Its percentile interval uses 10,000 hierarchical resamples of fresh processes, service templates and tenant sessions, with each process's measured service duration as denominator. It concerns this open-arrival trace, not the query-qualified closed-loop cohort.

S-VI. STRUCTURAL MEASUREMENT AND COST GRANULARITY

The main structural table concerns its six-source faithful cohort. Table S-XIX adds separately frozen five-family raw

TABLE S-XVI

ALL SIX PRIMARY SOURCES, WITH THREE PROCESS REPETITIONS NESTED WITHIN EACH SOURCE/MODEL. EACH ROW HAS 270 FUTURE PAIRS. SCORES USE THE FROZEN TASK ADAPTER; Δ IS THE ALL-ROUTE STRATEGY-MINUS-RAW MEAN. C DENOTES COMPILED PAIRS; POSITIVE CREDIT IS COUNTED WITHIN C. INCREASES/DECREASES USE $|\Delta s| > 0.03$. P99 IS RECOMPUTED OVER THAT ROW'S PAIRED FUTURE READINESS TIMES, INCLUDING QUALIFICATION WAITING; THESE ARE POINT ESTIMATES, NOT INDEPENDENT-SOURCE CONFIDENCE INTERVALS.

Source	Model	Raw mean	Δ mean	C / pairs	Raw > 0 / C	Up/down	p99 raw $\rightarrow$ CKV (ms)
H1	Qwen	0.500000	+0.000000	258/270	258/258	0/0	7,060.7 $\rightarrow$ 10,106.7
H1	Mistral	0.317460	+0.000000	258/270	258/258	0/0	7,616.7 $\rightarrow$ 10,137.5
H1	Llama	0.000000	+0.000000	90/270	0/90	0/0	6,349.6 $\rightarrow$ 8,363.4
H2	Qwen	0.140351	+0.000000	0/270	–	0/0	8,471.3 $\rightarrow$ 8,487.3
H2	Mistral	0.000000	+0.000000	84/270	0/84	0/0	9,303.1 $\rightarrow$ 15,000.6
H2	Llama	0.000000	+0.000000	90/270	0/90	0/0	7,860.2 $\rightarrow$ 10,265.5
H3	Qwen	0.500000	+0.000000	258/270	258/258	0/0	8,575.5 $\rightarrow$ 13,585.6
H3	Mistral	0.041331	+0.000000	0/270	–	0/0	9,432.5 $\rightarrow$ 9,491.8
H3	Llama	0.200000	+0.000000	90/270	90/90	0/0	7,859.9 $\rightarrow$ 10,064.7
R1	Qwen	0.320000	+0.000000	168/270	168/168	0/0	3,295.8 $\rightarrow$ 5,494.7
R1	Mistral	0.096667	+0.000000	0/270	–	0/0	4,632.0 $\rightarrow$ 4,628.0
R1	Llama	0.027370	+0.000000	0/270	–	0/0	3,245.9 $\rightarrow$ 3,239.8
R2	Qwen	0.281111	+0.000000	0/270	–	0/0	1,422.9 $\rightarrow$ 1,467.6
R2	Mistral	0.268889	+0.000000	0/270	–	0/0	1,803.2 $\rightarrow$ 1,807.6
R2	Llama	0.093333	+0.000000	0/270	–	0/0	1,410.6 $\rightarrow$ 1,401.8
R3	Qwen	0.113333	+0.000000	0/270	–	0/0	6,170.0 $\rightarrow$ 6,227.8
R3	Mistral	0.043333	+0.000000	0/270	–	0/0	8,061.9 $\rightarrow$ 8,087.7
R3	Llama	0.049704	+0.000519	0/270	–	5/3	6,006.7 $\rightarrow$ 5,989.7

TABLE S-XVII

NINE FROZEN CPU REGISTRATION-HELPER CASES: SEVEN REJECTS, ONE LEGITIMATE ACCEPTANCE AND ONE ACCEPTED SEMANTIC MISLABEL. THE HELPER VALIDATES DECLARATION AUTHORITY/CONSISTENCY, DOES NOT INFER CONFIDENTIALITY, AND IS NOT INTEGRATED INTO THE LIVE ENDPOINT.

Case	Result	Interpretation
Authorized operator template	Accept	Consistent authorized declaration
Tenant attempts shared registration	Reject	No operator registration authority
Forged operator identity	Reject	Issuer/tag mismatch
Changed bytes, stale hash	Reject	Content identity mismatch
Explicit tenant-private publication	Reject	Declared visibility conflicts with publication
Mutable segment marked for publication	Reject	Declared stability conflicts with publication
Missing authorization domain	Reject	Required registration scope absent
Version changed under old tag	Reject	Authenticated binding changed
Trusted operator mislabels private bytes	Accept	Valid declaration; confidentiality not inferred

TABLE S-XVIII

ARCHIVED QWEN DEPLOYMENT-TEMPLATE OPEN-ARRIVAL POINT: 25 TENANTS/TEMPLATE, 80% CALIBRATED LOAD, 500-MS SLO. GOODPUT UNITS ARE REQUESTS/S. THE 95% PERCENTILE INTERVAL RESAMPLES FRESH PROCESSES, SERVICE TEMPLATES AND TENANT SESSIONS HIERARCHICALLY (10,000 DRAWS); IT IS NOT A THREE-PROCESS MIN–MAX OR A CONFIDENCE INTERVAL ACROSS QUALIFICATION PROTOCOLS.

Raw	ContractKV	Increase	95% interval
0.172668	0.830651	0.657983	[0.538943, 0.765225]

TABLE S-XIX

BOUNDED FIVE-FAMILY RAW CONTROLS, SEPARATE FROM THE FAITHFUL COHORT: TEN SOURCES AND 270 FUTURE REQUESTS OVER THREE REPETITIONS PER MODEL/FAMILY. TOKEN-WEIGHTED PERCENTAGES.

Model	Family	Stable/all	Hits/all	Recovered
Qwen	RULER	75.679	6.786	1.313
Qwen	HotpotQA	88.075	3.756	0.267
Qwen	NarrativeQA	87.279	4.034	0.294
Qwen	Qasper	87.597	3.966	0.288
Qwen	RepoBench	91.417	1.770	0.000
Mistral	RULER	72.191	7.818	1.438
Mistral	HotpotQA	85.362	3.838	0.321
Mistral	NarrativeQA	84.190	4.221	0.358
Mistral	Qasper	84.738	4.112	0.347
Mistral	RepoBench	91.470	1.838	0.000
Llama	RULER	73.594	7.972	1.451
Llama	HotpotQA	87.432	4.217	0.278
Llama	NarrativeQA	87.006	4.651	0.295
Llama	Qasper	87.146	4.495	0.293
Llama	RepoBench	92.166	1.971	0.000

controls; Table S-XX holds four development sources fixed while changing input construction. Stable/all is stable body divided by all input tokens, Hits/all is total native hits divided by all input tokens, and Recovered is native hits on stable body divided by stable body tokens. Different cohorts and token attribution conventions are not treated as a before/after update of one aggregate. The token ledgers preserve actual prompt-ID alignment flags and separate source-macro from pooled token ratios.

TABLE S-XX
SAME FOUR DEVELOPMENT SOURCES, QWEN, 360 FUTURE REQUESTS/INPUT CONDITION, IDENTICAL 16K WINDOW AND DECLARED SERVICE PROTOCOL. PERCENTAGES USE THE SAME DEFINITIONS AS THE FIVE-FAMILY TABLE. THIS COMPARISON CHANGES INPUT CONSTRUCTION; IT IS NOT A COMPARISON BETWEEN THE ORIGINAL MOTIVATION AGGREGATES AND A NEW HELD-OUT POPULATION.

Input	Stable/all	Hits/all	Stable recovered
Word-cap packing	91.523	2.281	0.078
Structure-preserving	98.311	1.081	0.034

A. Matched Deployment Cost Components

For a trace with business mean $\bar{T}$, recorded verification durations c_j, slack s and normalization count N, residual accounting is $\bar{T} + \sum_j \max(c_j - s, 0)/N$. Setting $s = 0$ on the same recorded requests gives $\bar{T} + \sum_j c_j/N$. Table S-XXI preserves the original normalization and also gives an all-future denominator; these are component means, not new service-wall measurements. Qwen8B/Mistral originally normalize RPC cost by certified future requests; Qwen14B/Llama use all future requests. There are 1,350 future pairs/configuration, except Qwen14B with 675.

Qwen8B, Mistral and Qwen14B retain 147/147, 126/126 and 72/72 verifier-RPC timers, respectively. Three, 24 and three raw-reference generations instead reach their output limit before a verifier RPC is invoked. Their generation/controller elapsed times were not retained on the exception path. Llama records 75/108 RPC durations: rejected intermediate attempts retain decisions but lose timings. Full rejection also overwrites the cost with zero. Those missing costs are not imputed; the Llama savings are bounded above by the recorded-cost result. Missing preprocessing and raw-reference time prevent reconstructing complete historical controller cost. The independent registration-cost supplement and the primary directly measured service wall remain separate evidence.

TABLE S-XXI
MATCHED-REQUEST DEPLOYMENT ACCOUNTING: MEAN-TTFT REDUCTIONS (%) OVER THREE PROCESSES. RECORDED REPRODUCES RESIDUAL-SLACK ACCOUNTING; ZERO SLACK CHARGES RECORDED RPCS. LLAMA VALUES BOUND SAVINGS ABOVE BECAUSE 33 DURATIONS ARE MISSING. THESE ARE NOT COMPLETE CONTROLLER WALL OR MEASURED P99.

Configuration	Recorded	Zero/legacy	Zero/all
Qwen3-8B	37.81	16.81	28.91
Qwen3-14B TP2	44.69	34.39	34.39
Mistral-7B	31.72	2.83	19.74
Llama-3.1-8B	29.41	$\leq$20.37	$\leq$20.37

B. Where Binding Checks Execute

The source call chain places HMAC binding validation in the trusted deployment client before compiled/raw route selection, rather than in a separately deployed network gateway. Table S-XXII tests that function boundary. The domain-level certificate uses a wildcard tenant field; the application must authenticate users, enforce membership and mediate access to the engine. The separate manifest-helper cases in Table S-XVII test declaration consistency, not semantic confidentiality. No new GPU requests are used in these CPU audits.

TABLE S-XXII
CPU CHECKS OF THE ACTUAL DEPLOYMENT-CLIENT ISSUE/CHECK FUNCTION BODIES EXTRACTED FROM THEIR SOURCE. BINDING ERRORS DENY COMPILED SELECTION AND RETAIN RAW ROUTING; THESE CHECKS DO NOT IMPLEMENT A PUBLIC AUTHENTICATION PROXY OR MEASURE GPU PUBLICATION. THE TENANT-LABEL ACCEPTANCE IS EXPECTED UNDER THE DOMAIN WILDCARD AND REQUIRES APPLICATION-LEVEL USER AUTHORIZATION.

Condition	Cases	Compiled admissible
Matching domain binding	1	1
Changed bound identity/scope/settings	14	0
Missing binding, bad tag, expiry	3	0
Different tenant label; domain wildcard	1	1

REFERENCES

[1] L. Zheng *et al.*, "SGLang: Efficient execution of structured language model programs," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[2] I. Gim, G. Chen, S.-s. Lee, N. Sarda, A. Khandelwal, and L. Zhong, "Prompt Cache: Modular attention reuse for low-latency inference," in *Proc. Mach. Learn. Syst. (MLSys)*, vol. 6, 2024.

[3] J. Yao *et al.*, "CacheBlend: Fast large language model serving for RAG with cached knowledge fusion," in *Proc. 20th Eur. Conf. Comput. Syst. (EuroSys)*, 2025, doi: 10.1145/3689031.3696098.

[4] J. Hu *et al.*, "EPIC: Efficient position-independent caching for serving large language models," in *Proc. 42nd Int. Conf. Mach. Learn. (ICML)*, PMLR, vol. 267, 2025, pp. 24391–24402.

[5] C. Lin, Z. Han, C. Zhang, Y. Yang, F. Yang, C. Chen, and L. Qiu, "Parrot: Efficient serving of LLM-based applications with semantic variable," in *Proc. 18th USENIX Symp. Oper. Syst. Design Implement. (OSDI)*, 2024, pp. 929–945.

[6] C.-P. Hsieh *et al.*, "RULER: What's the real context size of your long-context language models?" in *Proc. Conf. Language Modeling (COLM)*, 2024.

[7] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning, "HotpotQA: A dataset for diverse, explainable multi-hop question answering," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2018, pp. 2369–2380, doi: 10.18653/v1/D18-1259.

[8] T. Kočiský *et al.*, "The NarrativeQA reading comprehension challenge," *Trans. Assoc. Comput. Linguistics*, vol. 6, pp. 317–328, 2018, doi: 10.1162/tacl_a_00023.

[9] P. Dasigi, K. Lo, I. Beltagy, A. Cohan, N. A. Smith, and M. Gardner, "A dataset of information-seeking questions and answers anchored in research papers," in *Proc. Conf. North American Chapter Assoc. Comput. Linguistics: Human Language Technologies (NAACL-HLT)*, 2021, pp. 4599–4610, doi: 10.18653/v1/2021.naacl-main.365.

[10] T. Liu, C. Xu, and J. McAuley, "RepoBench: Benchmarking repository-level code auto-completion systems," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[11] Y. Bai *et al.*, "LongBench: A bilingual, multitask benchmark for long context understanding," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL)*, 2024, pp. 3119–3137, doi: 10.18653/v1/2024.acl-long.172.

[12] A. Yang *et al.*, "Qwen3 technical report," *arXiv preprint arXiv:2505.09388*, 2025.

[13] A. Q. Jiang *et al.*, "Mistral 7B," *arXiv preprint arXiv:2310.06825*, 2023.

[14] A. Dubey *et al.*, "The Llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.

[15] vLLM Project, "Automatic prefix caching," *vLLM Documentation*, ver. 0.10.2. [Online]. Available: https://docs.vllm.ai/en/v0.10.2/design/prefix_caching.html. Accessed: Oct. 17, 2026.